

WARRENSBURG SKYHAVEN, MO, USA

WMO: 722367

Lat: 38.783N

Lon: 93.800W

Elev: 243

StdP: 98.44

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00363

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-12.1	-20.0	0.7	-13.9	-17.3	0.8	-10.4	10.5	4.5	9.4	1.2	2.6	290	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	34.8	22.9	32.6	23.7	31.4	23.5	26.0	30.7	25.2	30.3	24.4	29.3	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.6	29.1	23.0	18.3	27.9	22.6	17.9	27.5	82.3	30.1	77.9	30.1	74.6	29.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.3	8.3	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-19.7	27.2	4.3	1.1	-22.8	28.0	-25.3	28.7	-27.8	29.3	-30.9	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-1.0	1.3	7.8	12.5	18.0	23.4	24.7	23.7	20.2	13.5	7.0	1.3
	DBStd	10.28	6.33	6.36	6.18	5.21	4.66	3.28	3.15	3.07	4.14	5.18	5.48	5.50
	HDD10.0	1164	344	252	116	33	3	0	0	0	0	24	118	275
	HDD18.3	2728	599	477	329	186	66	4	1	2	27	167	341	529
	CDD10.0	2169	3	9	49	107	249	402	457	425	305	132	28	4
	CDD18.3	689	0	0	4	9	54	156	199	168	82	16	1	0
	CDH23.3	6446	0	1	27	83	434	1519	2014	1533	706	127	2	0
	CDH26.7	2409	0	0	1	11	103	570	842	611	247	24	0	0
Wind	WSAvg	3.1	3.5	3.8	3.9	3.9	3.1	2.9	2.2	2.1	2.3	3.1	3.5	3.5
Precipitation	PrecAvg	1063	34	49	71	118	142	154	117	85	110	86	63	46
	PrecMax	1422	93	125	160	310	313	286	355	225	282	207	209	115
	PrecMin	738	8	4	23	29	19	61	19	31	15	13	7	6
	PrecStd	203	21	30	36	54	79	61	76	46	82	46	50	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	22.2	26.2	27.8	31.3	35.9	37.9	36.4	33.8	29.7	22.7	18.8
		MCWB	12.7	14.5	16.7	18.4	22.3	23.3	22.0	24.2	22.5	21.2	15.6	15.2
	2%	DB	15.0	18.0	23.0	26.0	29.7	32.7	35.0	33.8	32.1	27.1	21.0	15.2
		MCWB	9.6	11.7	15.6	17.1	21.0	23.9	22.4	23.2	22.8	18.4	15.0	11.2
	5%	DB	11.3	15.3	21.1	23.0	27.5	32.1	32.6	32.2	30.0	24.1	17.8	12.3
		MCWB	6.9	10.3	14.5	15.8	20.3	23.7	24.0	23.5	22.5	17.0	12.9	8.9
	10%	DB	7.6	11.3	17.9	22.0	26.1	30.2	31.2	30.2	27.5	22.3	16.0	10.0
		MCWB	4.4	7.8	12.6	15.2	19.5	23.1	24.0	22.9	20.7	16.2	11.1	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	15.6	18.8	20.0	23.7	26.5	27.2	26.9	24.6	21.8	17.8	15.8
		MCDB	17.9	20.0	24.6	26.6	29.1	31.4	31.6	32.4	30.4	27.7	21.8	18.1
	2%	WB	10.0	13.7	17.3	18.7	22.6	25.5	26.3	25.6	23.5	20.3	15.7	12.4
		MCDB	13.9	17.4	22.4	23.9	27.6	30.6	30.4	30.9	29.7	24.2	18.8	14.6
	5%	WB	7.4	10.7	15.4	17.4	21.5	24.6	25.4	24.7	22.8	19.0	13.9	9.5
		MCDB	10.8	15.4	19.8	21.6	26.0	29.9	30.1	29.9	28.6	22.6	17.6	11.6
	10%	WB	4.6	7.6	13.3	15.8	20.3	23.6	24.5	23.7	22.0	17.4	11.8	6.7
		MCDB	7.9	11.3	17.4	20.3	24.6	28.8	29.0	28.5	26.7	20.6	15.1	9.5
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.0	10.4	11.2	10.5	10.7	10.5	10.9	11.5	12.0	10.8	9.0
		MCDBR	14.3	15.7	14.1	13.7	12.1	11.9	11.9	13.2	12.2	13.9	13.9	13.3
		MCWBR	9.6	10.6	7.6	7.6	5.9	5.1	4.2	4.9	5.0	6.8	9.0	9.7
	5% WB	MCDBR	13.5	15.1	12.2	11.3	10.7	10.6	9.8	11.0	11.1	11.1	11.9	12.4
MCWBR		9.7	11.1	7.5	7.4	6.0	5.4	5.0	5.1	5.2	6.5	9.1	10.5	
Clear-Sky Solar Irradiance	taub	0.297	0.310	0.337	0.389	0.428	0.429	0.444	0.446	0.389	0.347	0.307	0.292	
	taud	2.461	2.427	2.408	2.279	2.233	2.262	2.236	2.239	2.356	2.455	2.512	2.497	
	Ebn at Noon	883	917	922	888	856	852	835	825	856	862	866	864	
	Edh at Noon	80	95	107	130	139	135	138	134	112	92	76	72	
All-Sky Solar Radiation	RadAvg	2.18	2.92	3.83	4.91	5.68	6.43	6.42	5.66	4.80	3.42	2.37	1.88	
	RadStd	0.24	0.34	0.34	0.32	0.51	0.52	0.40	0.31	0.31	0.37	0.25	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page